Fashion Under Distortion: Robust MLP and CNN Classification with Prediction Consistency

Ali Hamza and Nabonita Das
Department of Artificial Intelligence
University of Michigan Dearborn
ECE 5831 — Final Project Proposal — Fall 2026

***Abstract*—We will develop fashion classifiers that tolerate noise, blur, and partial occlusion. Our proposed solution combines corruption augmentation with prediction-consistency regularization. We will compare multilayer perceptrons (MLPs) and convolutional neural networks (CNNs), isolate each training component, and quantify robustness, clean accuracy, and computational cost. The outcome will be a reproducible experimental pipeline and an interactive corruption demo.**

I. Problem and Prior Work

Clean-image accuracy can hide failures under degraded inputs. Fashion-MNIST [1] supplies 70,000 grayscale clothing images from ten classes. Inspired by corruption benchmarking [2] and AugMix's consistency principle [3], we ask whether penalizing disagreement between clean and corrupted predictions improves robustness beyond augmentation alone.

II. Data and Model Design

We will reserve 50,000 images for training and 10,000 for stratified validation from the official training split, preserving the 10,000-image official test set. Images are 28×28 pixels; transformations occur in $[0, 1]$ before normalization using training statistics. The MLP will use $784 \to 512 \to 64 \to 10$ units with ReLU hidden activations. The CNN will use two padded 3×3 convolution blocks with 32 and 64 filters, ReLU and 2×2 pooling, followed by 128 hidden units and ten logits. Their parameter counts will be approximately matched.

III. Proposed Solution and Evaluation

For each training image, we will sample one mild or moderate corruption: zero-mean Gaussian noise, Gaussian blur, or a black square occlusion. The full method will optimize

$$\mathcal{L} = \tfrac{1}{2}\big[\mathrm{CE}(p, y) + \mathrm{CE}(q, y)\big] + \lambda\, \mathrm{JS}(p, q), \qquad (1)$$

where p, q are clean and corrupted class probabilities, y is the label, CE is cross-entropy, and JS is Jensen–Shannon divergence. This encourages correct, consistent predictions. We will preregister $\lambda = 1$, Adam with learning rate 10^{-3}, batches of 128, and at most 30 epochs with validation-based early stopping.

Both architectures will use three training modes: clean-only, paired augmentation without JS, and paired augmentation with JS. Three seeds yield 18 final runs. We will freeze test corruptions at three severities per family, including a stronger level excluded from training, and reuse identical corrupted test images across methods. We will inspect representative distortions for label preservation; variants stay in their original partitions. We will report accuracy, macro-F1, clean-to-corrupted accuracy loss, parameters, training time, and mean/standard deviation across seeds. We hypothesize lower corruption error with consistency training; ablations will test this and quantify any clean-accuracy cost. Conclusions will concern these synthetic corruption families.

IV. Timeline and Individual Contributions

Oct. 23, 11:59 PM: submit proposal. **Oct. 24–Nov. 6:** prepare data and clean baselines. **Nov. 7–13:** prototype robust training and prepare a one-page progress update for the Canvas deadline. **Nov. 14–Dec. 4:** finish repeated experiments and analysis. **Dec. 5–14:** package deliverables by Dec. 14, 9:30 PM. Ali Hamza will lead the MLP and corruption generator; Nabonita Das will lead the CNN and consistency objective. Both will share evaluation, literature review, writing, and presentation.

V. GitHub Deliverables

The public `ece5831-2026-final-project` repository will contain versioned dependencies, configuration files, data preparation, model, training and evaluation scripts, seeds, checkpoints, results, a demo, and an executed `final-project.ipynb`. The README will give reproduction commands and link the dataset, slides, IEEE report, approximately 15-minute recorded presentation, and YouTube demo. Required files will also be organized in a Drive folder shared with anyone with the link. Both members will submit the repository URL individually.

References

[1] H. Xiao, K. Rasul, and R. Vollgraf, "Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms," *arXiv:1708.07747*, 2017.

[2] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *Proc. ICLR*, 2019.

[3] D. Hendrycks *et al.*, "AugMix: A simple data processing method to improve robustness and uncertainty," in *Proc. ICLR*, 2020.